

TITLE : Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Conclusion

This Java program successfully demonstrates the practical usage of the super keyword in class inheritance. By invoking parent class constructors via super(), the program ensures proper constructor chaining during object instantiation. Furthermore, utilizing super to access parent instance variables and call overridden parent methods prevents ambiguity when child classes share or modify parent behaviors. Masterfully leveraging super promotes clean code reusability, preserves parent initialization logic, and maintains a well-structured class hierarchy.

Exercise

1. Create an **Employee and Manager program** where Manager accesses Employee details using super keyword.

Code:

```
1 class Employee {
2     String name = "Alex Smith";
3     double baseSalary = 50000;
4     Employee(String name, double baseSalary) {
5         this.name = name;
6         this.baseSalary = baseSalary;
7     }
8     void displayInfo() {
9         System.out.println("Employee Name: " + name + " | Base Salary: $" + baseSalary);
10    }
11 }
12 class Manager extends Employee {
13     double bonus;
14     Manager(String name, double baseSalary, double bonus) {
15         super(name, baseSalary);
16         this.bonus = bonus;
17     }
18     @Override
19     void displayInfo() {
20         super.displayInfo();
21         System.out.println("Manager Bonus: $" + bonus + " | Total: $" + (super.baseSalary + bonus));
22     }
23     Run main | Debug main
24     public static void main(String[] args) {
25         Manager mgr = new Manager("Sarah Connor", 75000, 15000);
26         mgr.displayInfo();
27     }
28 }
```

Output:

```
Listening on 62080
User program running
Employee Name: Sarah Connor | Base Salary: $75000.0
Manager Bonus: $15000.0 | Total: $90000.0
User program finished
```

2. Develop a **Vehicle Insurance System** where child insurance classes access parent vehicle information using the super keyword.

Code:

```
1  class Vehicle {
2      String vehicleNumber;
3      double baseValue;
4      Vehicle(String vehicleNumber, double baseValue) {
5          this.vehicleNumber = vehicleNumber;
6          this.baseValue = baseValue;
7      }
8      void displayVehicleDetails() {
9          System.out.println("Vehicle No: " + vehicleNumber + " | Base Value: $" + baseValue);
10     }
11 }
12 class CarInsurance extends Vehicle {
13     double premiumRate = 0.05;
14     CarInsurance(String vehicleNumber, double baseValue) {
15         super(vehicleNumber, baseValue);
16     }
17     void calculateAndDisplayPremium() {
18         super.displayVehicleDetails();
19         double premium = super.baseValue * premiumRate;
20         System.out.println("Calculated Annual Premium: $" + premium);
21     }
22     Run main | Debug main
23     public static void main(String[] args) {
24         CarInsurance insurance = new CarInsurance("MH-12-AB-1234", 20000);
25         insurance.calculateAndDisplayPremium();
26     }
27 }
```

Output:

```
Listening on 56567
User program running
Vehicle No: MH-12-AB-1234 | Base Value: $20000.0
Calculated Annual Premium: $1000.0
User program finished
```